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UPPER AUSTRIA

» COURSE TYPE «

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» Title of your academic work can be very
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submitted as a » DOCUMENT TYPE «

to obtain the academic degree of

» degree «

by

» Authorfirstname Authorlastname, «

» Authorfirstname2 Authorlastname2 «

Wels, » submission month « » submission year «

Thesis supervisor

» Prof. Supervisor «



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» Prof. Supervisor «

Preface/Acknowledgments, Vorwort/Danksagung

The preface contains a short description of the topic; words of thanks to companies, mentors, supervisors, family members, etc., for their support are optional under consideration of relevant data protection terms.



Sworn declaration

I hereby declare that I prepared this work independently and without help from third parties, that I did not use sources other than the ones referenced and that I have indicated passages taken from those sources.

This thesis was not previously submitted in identical or similar form to any other examination board, nor was it published.

This printed thesis is identical with the electronic version submitted.

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.....

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Eidesstattliche Erklärung

Ich erkläre eidesstattlich, dass ich die vorliegende Arbeit selbstständig und ohne fremde Hilfe verfasst, andere als die angegebenen Quellen nicht benutzt und die den benutzten Quellen entnommenen Stellen als solche kenntlich gemacht habe.

Die Arbeit wurde bisher in gleicher oder ähnlicher Form keiner anderen Prüfungsbehörde vorgelegt und auch nicht veröffentlicht.

Die vorliegende Arbeit ist mit dem elektronisch übermittelten Textdokument identisch.

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Abstract

The Abstract should meet the following criteria:

- Complete summary of the thesis in **English**
- Not longer than one page
- No description of complicated facts
- Clear definition of the problem, objectives, approach, methods used, results and solutions

Kurzfassung

Die Kurzfassung soll folgende Kriterien erfüllen:

- Verständliche Zusammenfassung der Arbeit auf **Deutsch**
- Umfang von maximal einer Seite
- Keine komplizierten Sachverhaltsdarstellungen
- Klare Herausarbeitung der Problemstellung, Zielsetzungen, Vorgehensweise, angewandten Methoden, Ergebnisse und Problemlösungen

Contents

Preface/Acknowledgments, Vorwort/Danksagung	III
Sworn declaration	IV
Eidesstattliche Erklärung	V
Abstract	VI
Kurzfassung	VII
1 Introduction	1
1.1 Headings	2
1.2 Dashes	3
1.3 Numbered and unnumbered lists	4
1.3.1 Itemize	4
1.3.2 Enumerate	4
1.3.3 Outline package	5
1.4 Insert a table	6
1.5 Longtable and importing data	8
1.5.1 Table from an Imported CSV (DTL)	8
1.5.2 Longtable example	8
1.6 Figures	11
1.6.1 Multiple figures in one environment	11
1.6.2 Usage of the overpic package	12
1.6.3 Usage of wrapfig package	14
1.6.4 On using figures	14
1.7 Mathematics and Units	15
1.7.1 Variables and Constants	15
1.7.2 Mathematical fonts	15
1.7.3 Sub and superscript	16
1.7.4 Physics package	16
1.7.4.1 Differentials	17
1.7.5 Greek Symbols in Text	17
1.7.6 The <code>siunitx</code> Package	17
1.7.7 Custom commands	18
1.7.7.1 Number spaces	18
1.7.7.2 Umlauts in math mode	18
1.7.7.3 Dimensionless numbers	19
1.7.7.4 Abbreviations	19

2	PGF and TikZ	21
2.1	Baby's first <code>tikzpicture</code>	21
2.2	Flowchart with TikZ	21
2.3	Diagrams with PGFPlots	22
2.4	Plotting External Data	23
2.5	Distinct Line Styles	24
2.6	Using PGF for tables	25
3	Theoretical observation	26
4	Bibliography	27
5	List of Figures	28
6	List of Tables	29
A	Included PDF files	30
B	Source Code	34
B.1	Markup and data languages	34
B.1.1	L ^A T _E X code example	34
B.1.2	YAML code example	34
B.2	Programming languages	34
B.2.1	Python code example	34
B.2.2	Matlab code example	35
B.2.3	Wolfram Mathematica code example	35
B.2.4	C code example	35
B.2.5	Longer code example (C)	37

1 Introduction

Some experience with the software package $\text{\LaTeX}$ is necessary for using this template. Please refer to the `README.MD` file or the GitHub repository for installation and first compilation.

Throughout this template, various informational boxes are used to communicate important details, recommendations, and warnings. Each type of box serves a distinct purpose:

NOTE

Provides general notes or remarks. These highlight information that users should be aware of, even when skimming the document.

TIP

Contains optional advice or best practices. These tips are intended to improve the user's workflow or output but are not strictly required.

IMPORTANT

Marks essential information that is critical for the correct usage of the template. Ignoring these instructions may lead to incorrect formatting, compilation issues, or failure to meet submission requirements.

DANGER

Indicates critical issues that require immediate attention. Failure to comply may result in severe errors, document corruption, or non-compliance with institutional standards.

WARNING

Highlights potential risks, unintended consequences, or problematic outcomes that could arise from a particular action or configuration.

QUIRK

Documents minor irregularities, limitations, or edge cases related to $\text{\LaTeX}$ behavior, package interactions, or the template itself. These do not generally prevent successful use but may cause confusion or unexpected results.

🐛 BUG

Reports known issues, malfunctions, or broken features in the current version of the template. These require manual workarounds or await future fixes.

You can disable the feature in the modules. Simply comment out `\input{preamble/mod-callout}`. Keep in mind that you will need to also disable any files that use the feature

1.1 Headings

With respect to headings, consider the following points for the formulation of a table of contents:

- Do not use unknown abbreviations in headings
- No full stop at the end of headings
- Make sure that a heading – without associated text – is not situated on the last line of a page; a page should also not start with a single line which belongs to the paragraph on the previous page.
- The degree of detail corresponds to the focus and emphasis of the thesis
- Always use at least two sub-chapters in one chapter; otherwise do not use sub-chapters
- Use a maximum of 5 levels of sub-chapters

The following are the available heading types. The numbering is identical to the internal $\text{\LaTeX}$ numbering. You can add an asterisk (*) in to remove the number and the table of contents entry. Use `\addchap` or `\addsec` for unnumbered headings with a table of contents entry.

level 0: <code>\chapter{...}</code>	<code>\addchap{...}</code>
level 1: <code>\section{...}</code>	<code>\addsec{...}</code>
level 2: <code>\subsection{...}</code>	not provided
level 3: <code>\subsubsection{...}</code>	not provided
level 4: <code>\paragraph{...}</code>	not numbered
level 5: <code>\subparagraph{...}</code>	not numbered

The numbering for paragraph and subparagraph has been disabled and no table of contents entry will be created.

1.2 Dashes

Hyphen (-) (–)

The shortest dash. Used for:

- Hyphenating words (e.g., well-being, co-author)
- Breaking words at line ends
- Compound modifiers (e.g., high-speed train)

En dash (–) (—)

Medium length dash. Use it for:

- Ranges (e.g., 1990–2025, pages 12–30)
- Connecting related items or contrast (e.g., the Paris–London flight)
- Scores or results (e.g., 3–2 win)

Em dash (—) (— — —)

Longest dash. Used to:

- Indicate breaks or interruptions in thought
- Emphasize additional information or explanations
- Replace commas, parentheses, or colons for emphasis

Minus sign (−)

Not a dash, but often confused with one. Used in math to denote subtraction or negative numbers. In L^AT_EX, use a normal hyphen in math mode for a proper minus.

1.3 Numbered and unnumbered lists

This serves as a demonstration of the `itemize` and `enumerate` environments and their possibilities.

WARNING: `\subitem` is not used for lists, it is a command used in indices.

Note: The spacing of the items has been reduced compared to the default value for higher density of the text.

1.3.1 Itemize

- First level item 1
- First level item 2
- First level item 3
- First level item 4
- First level item 5

You can use `\begin{itemize} ... \end{itemize}` within itself (nesting) to create sub items.

Note: In this document, `itemize` lists use horizontal rules in place of default bullets, with size decreasing by level.

- Level 1 item
 - Level 2 item
 - Level 3 item
 - Level 4 item

1.3.2 Enumerate

1. First level item 1
2. First level item 2
3. First level item 3
4. First level item 4
5. First level item 5

Here is an example of a nested `enumerate` environment. `enumerate` lists have been customized to show hierarchical numbering: 1., 1.1., 1.1.1., etc.

1. First level item 1
 - 1.1. Second level item 1
 - 1.1.1. Third level item 1
 - 1.1.1.1. Fourth level item 1

1.3.3 Outline package

This example uses the `outlines` package (note the plural). A separate and unrelated `outline` package also exists – be sure to refer to the correct one.

The customization done via `enumitem` carry over to the outline environment.

- Item on level 1
 - Item on level 2
 - Item on level 3
 - Item on level 4

Here is what that looks like in the code:

```
\begin{outline}
  \1 Item on level 1
  \2 Item on level 2
  \3 Item on level 3
  \4 Item on level 4
\end{outline}
```

The `\1` etc. provide the "level" indicator for the list.

Here is an `outline` environment using the `[enumerate]` option right after `\begin{outline}`.

1. Top level item
 - 1.1. Sub item
 - 1.1.1. sub sub item

You can use a one-time custom label with square brackets, e.g. `\1[>]`. The default label (line or number, depending on choice) is overwritten.

- > a very simple example using the normal greater sign
 - you can use the math commands, here: `[$\rightarrowtail$]`
- L^AT_EX be creative
 - or don't

fixed right: yes, all of that is the label. Be careful about the left margin!

1.4 Insert a table

This section demonstrates how to create tables using different environments and styling options.

Tables in $\text{\LaTeX}$ are highly customizable but can be verbose. The examples below show basic layouts, row colouring, automatic column width, and data import from external files.

Simple Table with Rulings

This example shows a standard table with manual horizontal and vertical lines. Different ruling styles (`|`, `\hline`, `\headrule`) help visually separate headers and rows. This is the most basic way to create tables.

Note: Manual layout like this is error-prone for complex tables or when column widths need to adapt.

Tab. 1.1: Simple Table with Rulings

Heading column 1	Heading column 2	Heading column 3
1	Test-1	Test-2
2	0	0
3	0	0
4	0	1

Table with Alternating Row Colors

Here, the `\rowcolors` command is used to improve readability by applying alternating row colors (zebra striping). This technique is particularly helpful for tables with many rows.

However, the columns use fixed-width definitions (`p{3cm}`), which allows text wrapping but must be set manually.

Tab. 1.2: Table with alternating row colours

size	size 1	size 2
[unit]	[unit 1]	[unit 2]
object 1	5	6
object 2	8	9
object 3	11	12

Table with Automatic Column Width — Tabularx

The `tabularx` package automatically adjusts column widths to fit the table into a specified width (here, 80% of the text width). This reduces manual tweaking but is less compatible with features like `\rowcolors` or certain advanced table rules.

Columns labelled X auto-expand.

Mixed with standard c or fixed-width p columns.

Tab. 1.3: Table with alternating colours and automatic column width (B) using `tabularx`, fixed size (C) and standard centred (A)

A	B	B	C
1	2	2	3
4	5	5	6
7	8	8	9
10	11	11	12

Table with Justified Columns — Tabulary

The `tabulary` package extends `tabularx` by offering precise control over text alignment within columns. It supports the following alignment options:

- **L** ↪ Left aligned (equivalent to X)
- **C** ↪ Centred
- **R** ↪ Right aligned
- **J** ↪ Fully justified text

This is particularly useful for tables that combine descriptive text with numerical data, allowing for readable layouts without manual width adjustments.

i NOTE

Compatibility with `\rowcolors` and very wide tables is limited. Be cautious when mixing with automatic row coloring or complex layouts.

Tab. 1.4: Example using `tabulary`, smaller text to save space

Left-aligned	Centred	Right-aligned	Justified text
Apple	10	5.99	This is some longer text that will wrap properly.
Banana	20	3.49	Another line with text to test justification.
Cherry	15	7.99	Yet another example.

1.5 Longtable and importing data

This section shows how to create tables that automatically split across multiple pages and how to import data from external files like CSVs.

💡 TIP

See also section 2.6 for a different solution to data importing.

1.5.1 Table from an Imported CSV (DTL)

This table is generated directly from a CSV file using the `datatool` package. This avoids manual data entry in LaTeX and ensures consistency if the data updates.

Useful for:

- Simple data tables
- Automatically generated or exported datasets (e.g., from Excel)

Tab. 1.5: Table from an imported CSV

Col1	Col2	Col3	ColPct
111	65	64	5%
112	30	5	6%
113	92	1	8.4%
114	47	19	20%
115	38	15	1%

1.5.2 Longtable example

When tables are too long to fit on one page, `longtable` handles automatic splitting with repeated headers. Combined with `datatool`, this creates dynamic, multi-page tables.

This example loads a CSV file (`sales_data.csv`) and displays the data with:

- Repeated headers on each page
- Alternating row colors
- Custom column widths

This method is robust but comes with more complex syntax.

 TIP

External data import works best for datasets that might change frequently or for long reports.

It can be useful to place the code for these imports in a separate file for better readability and reusability

 QUIRK

`longtable` does not work inside floating environments like `table`.

Tab. 1.6: Longtable from a CSV file

Date	Product	Units Sold	Price (€)	Total (€)
2024-01-01	Laptop	5	1200	6000
2024-01-02	Smartphone	8	800	6400
2024-01-03	Tablet	3	450	1350
2024-01-04	Headphones	12	150	1800
2024-01-05	Monitor	6	300	1800
2024-01-06	Keyboard	7	100	700
2024-01-07	Mouse	10	50	500
2024-01-08	Smartwatch	4	250	1000
2024-01-09	Printer	2	200	400
2024-01-10	Router	9	180	1620
2024-01-11	Laptop	5	1200	6000
2024-01-12	Smartphone	8	800	6400
2024-01-13	Tablet	3	450	1350
2024-01-14	Headphones	12	150	1800
2024-01-15	Monitor	6	300	1800
2024-01-16	Keyboard	7	100	700
2024-01-17	Mouse	10	50	500
2024-01-18	Smartwatch	4	250	1000
2024-01-19	Printer	2	200	400
2024-01-20	Router	9	180	1620
2024-01-21	Laptop	5	1200	6000
2024-01-22	Smartphone	8	800	6400
2024-01-23	Tablet	3	450	1350

(Continued on next page)

Continued from previous page(opt.)

Date	Product	Units Sold	Price (€)	Total (€)
2024-01-24	Headphones	12	150	1800
2024-01-25	Monitor	6	300	1800
2024-01-26	Keyboard	7	100	700
2024-01-27	Mouse	10	50	500
2024-01-28	Smartwatch	4	250	1000
2024-01-29	Printer	2	200	400
2024-01-30	Router	9	180	1620
2024-01-31	Laptop	5	1200	6000
2024-02-01	Smartphone	8	800	6400
2024-02-02	Tablet	3	450	1350
2024-02-03	Headphones	12	150	1800
2024-02-04	Monitor	6	300	1800
2024-02-05	Keyboard	7	100	700
2024-02-06	Mouse	10	50	500
2024-02-07	Smartwatch	4	250	1000
2024-02-08	Printer	2	200	400
2024-02-09	Router	9	180	1620
2024-02-10	Laptop	5	1200	6000
2024-02-11	Smartphone	8	800	6400
2024-02-12	Tablet	3	450	1350
2024-02-13	Headphones	12	150	1800
2024-02-14	Monitor	6	300	1800
2024-02-15	Keyboard	7	100	700
2024-02-16	Mouse	10	50	500
2024-02-17	Smartwatch	4	250	1000
2024-02-18	Printer	2	200	400
2024-02-19	Router	9	180	1620

1.6 Figures

Figures in $\text{\LaTeX}$ are handled using the `figure` environment. Like tables, figures are inserted as floats. This means they are positioned where $\text{\LaTeX}$ deems optimal for page layout, rather than exactly where they are written in the code. By default, floats tend to move to the top or bottom of pages or onto dedicated float pages.

To force figures closer to their position in the code, use placement specifiers:

- `[h]` — roughly "here"
- `[H]` — exactly "here" (requires the `float` package)

It is best practice to scale images relative to the text width, e.g., `width=0.5\textwidth`, to maintain consistency across different document sizes.



Fig. 1.1: Example of inserting a figure

1.6.1 Multiple figures in one environment

The simplest way to display two images side by side is to call `\includegraphics` twice inside one `figure` environment. However, spacing must be managed manually.

Example with manual spacing:



Fig. 1.2: Example of inserting two figures

`\hfil` provides flexible horizontal spacing and is generally the best choice.

Alternatives include `\hspace`, `\hskip`, and explicit spacing lengths.

Using `subfigure` (Recommended)

A more robust method uses the `subcaption` package. This allows each image to have its own caption and label for referencing.

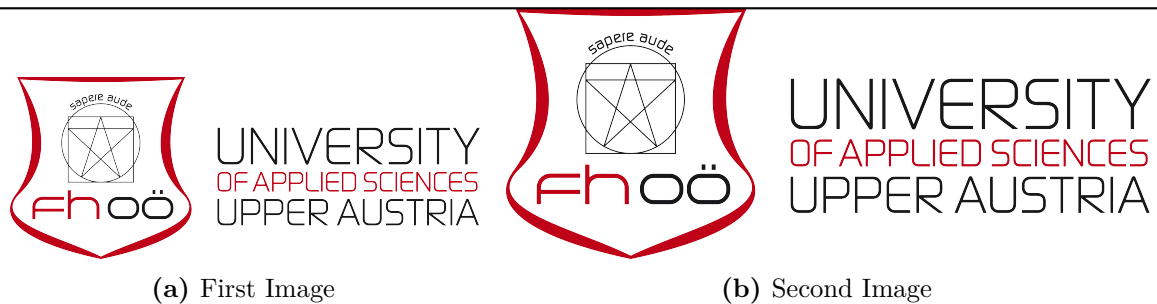


Fig. 1.3: Example of inserting a figure with differently sized subfigures with subcaption

⚠ WARNING

A `subfigure` package also exists. It is incompatible with `subcaption` and is less versatile.

The widths of subfigures should sum to less than or equal to 1.0 relative to `\textwidth` or `\linewidth`, accounting for spacing. Typical choices are `0.49\linewidth` or `0.33\linewidth`.

i NOTE

There is a difference between `\textwidth` and `\linewidth`.

`\textwidth` refers to the total width of the main text block on the page. It stays constant in single-column documents.

`\linewidth` refers to the width of the current line of text — which changes depending on the environment. Inside a figure, table, or other floats, `\linewidth` is equal to `\textwidth`.

However, inside nested environments like `minipage`, `subfigure`, `quote`, or `wrapfigure`, `\linewidth` adjusts to the local width.

This means that when sizing images inside subfigures or wrapped figures, you should generally use `width=\linewidth` rather than `\textwidth` to ensure the image fills the container correctly.

1.6.2 Usage of the `overpic` package

The `overpic` package is too big to be explained here. Please refer to its own documentation. You can use the `grid` option to help with positioning.

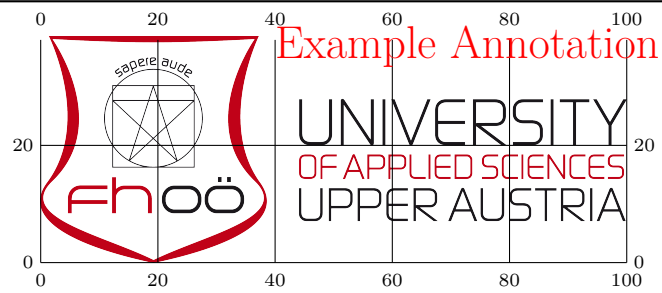


Fig. 1.4: Example using the overpic package

1.6.3 Usage of wrapfig package

This is an example of a paragraph with a wrapped figure. The text will flow around the image automatically. You can adjust the figure width, placement, and spacing to optimize the layout. If the text is too short, you might see issues with overlapping or strange formatting. This is an example of a paragraph with a wrapped figure. The text will flow around the image automatically. You can



Fig. 1.5: A wrapped figure with reduced spacing.

adjust the figure width, placement, and spacing to optimize the layout. If the text is too short, you might see issues with overlapping or strange formatting. This is an example of a paragraph with a wrapped figure. The text will flow around the image automatically. You can adjust the figure width, placement, and spacing to optimize the layout. If the text is too short, you might see issues with overlapping or strange formatting.

The spacing of the image has been adjusted to move it further upwards.

1.6.4 On using figures

Consider the following points when using figures:

- Figures must be referenced within or linked to the text; otherwise they are superfluous. To do this, use the command `\cref{}` [see fig. 1.1]
An additional page reference can be added with the command `\pageref{}` [see fig. 1.1 on page 11]
- Not every figure needs to be in the text. Analyses, tables, backup information, etc. can be collected in the appendix. However, the text should reference them.
- If necessary, add a legend in the figures and the necessary references.

1.7 Mathematics and Units

Writing mathematical equation in $\text{\LaTeX}$ is generally easy, the problem lies in making them look correct. There are many common practices that are specific to $\text{\TeX}$ users, scientific writing in general, and can be very different depending of the field of study. This template focuses on the usage for mechanical & electrical engineering for central Europe.

1.7.1 Variables and Constants

It is common practice to write variables in *italics* and constants upright. For example, k may be the stiffness of a spring but would be typeset as k (though more commonly k_B) for the Boltzmann constant.

Keep in mind that the usage of these principles also applies to Greek letters for example: α and α , or ϵ and ε . Be aware of π and ϖ especially, ϖ (the second one) looks more like a lower case omega, than π .

NOTE

Euler's constant e was formatted in `\mathscr{}` instead of upright (`\mathrm{}`) for better usability in text and aesthetic purposes

TIP

For more examples, please see the following repositories:

<https://github.com/Gugarutz/FH-Wels-Formelsammlung-THD-STL-WUE>

<https://github.com/Gugarutz/FH-Wels-FEL-Formelsammlung>

1.7.2 Mathematical fonts

The use of fonts (or rather typefaces) in $\text{\LaTeX}$ can be hard to understand, especially due to the misuse of the word by Microsoft Office products. Fonts are variations of a single typeface, eg. bold, italics, or small caps. Whereas typefaces are what you may know as fonts.

Keep in mind that the typefaces for math and text are always separated in $\text{\LaTeX}$, both in regards of setting the defaults and in terms of availability.

Command	Text	Numbers	Description
Default (none)	<i>ABC</i>	123456789.0	Default math italic
<code>\mathrm{...}</code>	ABC – abc	123456789.0	Upright Roman (for constants)
<code>\mathit{...}</code>	<i>ABC – abc</i>	<i>123456789.0</i>	Italic (for variables)
<code>\mathbf{...}</code>	ABC – abc	123456789.0	Bold Roman (serif)
<code>\mathsf{...}</code>	ABC – abc	123456789.0	Sans-serif upright
<code>\mathtt{...}</code>	ABC – abc	123456789.0	Typewriter (monospace)
<code>\mathcal{...}</code>	<i>ABC</i>	—	Calligraphic (uppercase only)
<code>\mathbb{...}</code>	<i>ABC</i>	—	Blackboard (uppercase only)
<code>\mathds{...}</code>	ABC	1.	alternate Blackboard (uppercase only)
<code>\mathfrak{...}</code>	<i>ABC – abc</i>	<i>6789.0</i>	Fraktur (Gothic-style)
<code>\boldsymbol{...}</code>	ABC – abc	123456789.0	General bold math (symbols & Greek)
<code>\mathscr{...}</code>	<i>ABC</i>	—	Script (more decorative)

1.7.3 Sub and superscript

Normal subscript and superscript comes after the object in question and is done with `_` and `^`. By default this only applies to one symbol each, so `M_ab` produces M_ab . To avoid this use curly brackets like with other $\text{\LaTeX}$ commands as follows: `M_{ab}^{cd}` to produce M_{ab}^{cd} . Also avoid M_{a^b} (`M_{a^b}`) and M_a^b (`{M_a}^b`).

With the `\prescript{a}{b}{M}` command you can typeset subscript and superscript before an object like this:

$${}_b^a M \quad \text{and even add the ones after that:} \quad {}_b^a M_c^d$$

1.7.4 Physics package

The `physics` package defines `\qty`. However this is already used by `siunitx`. Therefore this template redefines `\qty` to `\set`.

`\set{10}` produces: $\{10\}$

`\set[\dfrac{13}{14}]` produces: $\left[\frac{13}{14}\right]$

`\set|\vec{a}|` `\abs{\vec{r}}` `\norm{\va{r}}` `\norm{\va*{r}}`

produces:

$$|\vec{a}| \quad |\vec{r}| \quad \|\vec{r}\| \quad \|\vec{r}\|$$

WARNING

`\set` may only be used inside a math environment!

1.7.4.1 Differentials

The `physics` package adds various helpful commands for easy and quick setting of differentials.

The normal upright `d` used in by `physics` was replaced by `dL` which is used by professor Schiefermayr in his scripts. You can disable this in the `preamble/math.tex` under the custom math commands.

$$\int_0^\infty 3x^2 \, dLx = x^3 \Big|_0^\infty$$

1.7.5 Greek Symbols in Text

With the `textgreek` package, you can use commands like `\textalpha` (α) to create Greek characters inside normal text without needing to invoke math mode.

1.7.6 The `siunitx` Package

This package is useful for automatic typesetting of numbers and units. You will need to choose appropriate settings in `\unitsetup{...}` in the `main.tex` preamble.

Only Units

The `\unit` command ensures proper spacing and styling for standalone units:

- `\unit{\kilogram\metre\per\second}` produces kg m s^{-1} .
- `\unit{\kg\per\m\per\s\squared}` produces $\text{kg m}^{-1} \text{s}^{-2}$.
- `\unit{\degreeCelsius}` correctly handles special symbols: $^{\circ}\text{C}$.
- `\unit{\ohm}` ensures correct typesetting of units like ohms: Ω .

Only Numbers

The `\num` command formats numbers consistently:

- `\num{12345.678}` formats large numbers: 12 345.678.
- `\num{6.022e23}` correctly formats scientific notation: $6.022 \cdot 10^{23}$.
- `\num{000.000123}` removes unnecessary leading zeros: 0.000 123.

Numbers and Units

The `\qty` command combines numbers and units seamlessly:

- `\qty{9.81}{\metre\per\second\squared}` represents acceleration: 9.81 m s^{-2} .
- `\qty{1.5}{\mega\hertz}` correctly applies SI prefixes: 1.5 MHz.
- `\qty{500}{\milli\liter}` handles subunits properly: 500 mL.

Uncertainty

- `\num{1.23(45)}` shows uncertainty in parentheses: 1.23 ± 0.45 .
- `\qty{1.23(45)}{\metre}` combines uncertainty with units: $(1.23 \pm 0.45) \text{ m}$.
- `\num{1.23(4)e5}` formats numbers with uncertainty in scientific notation: $(1.23 \pm 0.04) \cdot 10^5$.

To display uncertainty separately, use:

- `\qty[separate-uncertainty=false]{1.23(0.05)}{\metre}`,
which produces $1.23(5) \text{ m}$.

Display Options

Different display options for units and numeric notation can be configured in `\unitsetup{...}` globally or locally.

The `\num` command can be forced into scientific notation using the `scientific-notation=true` option:

- `\num[exponent-mode=scientific]{000.000123}` produces $1.23 \cdot 10^{-4}$.

The `\unit` and `\qty` command can format units as fractions:

- `\unit[per-mode=fraction]{\kg\per\m\per\s\squared}` produces $\frac{\text{kg}}{\text{m s}^2}$.
- `\unit[per-mode=symbol]{\kg\per\m\per\s\squared}` produces $\text{kg}/(\text{m s}^2)$.

1.7.7 Custom commands

Some special mathematical characters are provided by this template in a nicer and distinct font or typeset upright as per section 1.7.1.

Euler's constant: `\e` $\rightarrow e$

Imaginary i: `\imag` $\rightarrow i$

Imaginary j: `\jmag` $\rightarrow j$

1.7.7.1 Number spaces

The following commands are provided number sets:

`\N` $\rightarrow \mathbb{N}$ `\Z` $\rightarrow \mathbb{Z}$ `\Q` $\rightarrow \mathbb{Q}$ `\R` $\rightarrow \mathbb{R}$ `\I` $\rightarrow \mathbb{I}$ `\C` $\rightarrow \mathbb{C}$

1.7.7.2 Umlauts in math mode

Some extra math shortcuts are provided for German users for Umlauts as the normal ä, ö, ü would be typeset upright and require brackets `{ }` despite being one letter.

$$\begin{array}{lll}\backslash\mathrm{aum} \rightarrow \ddot{a} & \backslash\mathrm{oum} \rightarrow \ddot{o} & \backslash\mathrm{uum} \rightarrow \ddot{u} \\ \backslash\mathrm{Aum} \rightarrow \ddot{A} & \backslash\mathrm{Oum} \rightarrow \ddot{O} & \backslash\mathrm{Uum} \rightarrow \ddot{U}\end{array}$$

Compared to normal Umlauts being used in math mode:

$$\begin{array}{lll}P_{\ddot{a}} \rightarrow P_{\ddot{a}} & T_{\ddot{o}} \rightarrow T_{\ddot{o}} & p_{\ddot{u}} \rightarrow p_{\ddot{u}} \\ P_{\ddot{A}} \rightarrow P_{\ddot{A}} & T_{\ddot{O}} \rightarrow T_{\ddot{O}} & p_{\ddot{U}} \rightarrow p_{\ddot{U}}\end{array}$$

1.7.7.3 Dimensionless numbers

This template provides a significant set of preset named, dimensionless numbers. Be aware that these are not simple shortcuts but have specific kerning (the distance between characters) for better readability and recognizability.

The commands can be found in `preamble/mod-math.tex` file, you may disable them there. See table below for all available shortcuts/commands.

Tab. 1.7: All dimensionless numbers provided

<code>\Arch</code> $\rightarrow$ <i>Ar</i>	<code>\Biot</code> $\rightarrow$ <i>Bi</i>	<code>\Cauc</code> $\rightarrow$ <i>Ca</i>	<code>\Damk</code> $\rightarrow$ <i>Da</i>
<code>\Ecke</code> $\rightarrow$ <i>Ec</i>	<code>\Eule</code> $\rightarrow$ <i>Eu</i>	<code>\Four</code> $\rightarrow$ <i>Fo</i>	<code>\Frou</code> $\rightarrow$ <i>Fr</i>
<code>\Gras</code> $\rightarrow$ <i>Gr</i>	<code>\Karl</code> $\rightarrow$ <i>Ka</i>	<code>\Knud</code> $\rightarrow$ <i>Kn</i>	<code>\Lewi</code> $\rightarrow$ <i>Le</i>
<code>\Mach</code> $\rightarrow$ <i>Ma</i>	<code>\Nuss</code> $\rightarrow$ <i>Nu</i>	<code>\Pec1</code> $\rightarrow$ <i>Pe</i>	<code>\Pran</code> $\rightarrow$ <i>Pr</i>
<code>\Ray1</code> $\rightarrow$ <i>Ra</i>	<code>\Reyn</code> $\rightarrow$ <i>Re</i>	<code>\Schm</code> $\rightarrow$ <i>Sc</i>	<code>\Sher</code> $\rightarrow$ <i>Sh</i>
<code>\Stan</code> $\rightarrow$ <i>St</i>	<code>\Stro</code> $\rightarrow$ <i>Sr</i>	<code>\Thom</code> $\rightarrow$ <i>Th</i>	<code>\Webe</code> $\rightarrow$ <i>We</i>

1.7.7.4 Abbreviations

to be redone

Thermodynamics

<code>\eff</code> $\rightarrow$ <i>eff</i> Effective	<code>\inl</code> $\rightarrow$ <i>in</i> Inlet	<code>\out</code> $\rightarrow$ <i>out</i> Outlet
<code>\refe</code> $\rightarrow$ <i>ref</i> Reference	<code>\sat</code> $\rightarrow$ <i>sat</i> Saturation	<code>\amb</code> $\rightarrow$ <i>amb</i> Ambient
<code>\env</code> $\rightarrow$ <i>env</i> Environment	<code>\crit</code> $\rightarrow$ <i>crit</i> Critical	<code>\tot</code> $\rightarrow$ <i>tot</i> Total
<code>\rev</code> $\rightarrow$ <i>rev</i> Reversible	<code>\irr</code> $\rightarrow$ <i>irr</i> Irreversible	<code>\ex</code> $\rightarrow$ <i>ex</i> Exergy

numerics

<code>\avg</code> $\rightarrow$ <i>avg</i> Average	<code>\maxx</code> $\rightarrow$ <i>max</i> Maximum	<code>\minn</code> $\rightarrow$ <i>min</i> Minimum
<code>\iter</code> $\rightarrow$ <i>iter</i> Iteration	<code>\res</code> $\rightarrow$ <i>res</i> Residual	<code>\number</code> $\rightarrow$ <i>num</i> Numerical
<code>\expmt</code> $\rightarrow$ <i>exp</i> Experimental	<code>\simu</code> $\rightarrow$ <i>sim</i> Simulated	<code>\init</code> $\rightarrow$ <i>init</i> Initial
<code>\final</code> $\rightarrow$ <i>final</i> Final	<code>\CFL</code> $\rightarrow$ <i>CFL</i> Courant–Friedrichs–Lewy	

Thermal

<code>\thrm</code> $\rightarrow$ <i>th</i> Thermal	<code>\cond</code> $\rightarrow$ <i>cond</i> Conduction	<code>\conv</code> $\rightarrow$ <i>conv</i> Convection
<code>\rad</code> $\rightarrow$ <i>rad</i> Radiation	<code>\gen</code> $\rightarrow$ <i>gen</i> Generated	<code>\loss</code> $\rightarrow$ <i>loss</i> Loss
<code>\trans</code> $\rightarrow$ <i>trans</i> Transferred		

chem

<code>\eqlb</code> $\rightarrow$ <i>eq</i> Equilibrium	<code>\rxn</code> $\rightarrow$ <i>rxn</i> Reaction	<code>\reac</code> $\rightarrow$ <i>reac</i> Reaction (alt)
<code>\mix</code> $\rightarrow$ <i>mix</i> Mixture	<code>\bulk</code> $\rightarrow$ <i>bulk</i> Bulk	<code>\wall</code> $\rightarrow$ <i>wall</i> Wall
<code>\film</code> $\rightarrow$ <i>film</i> Film	<code>\surf</code> $\rightarrow$ <i>surf</i> Surface	<code>\mol</code> $\rightarrow$ <i>mol</i> Molecular Mass

fluids

<code>\visc</code> $\rightarrow$ <i>visc</i> Viscous	<code>\incomp</code> $\rightarrow$ <i>incomp</i> Incompressible
<code>\comp</code> $\rightarrow$ <i>comp</i> Compressible	<code>\ideal</code> $\rightarrow$ <i>ideal</i> Ideal (Gas)
<code>\realg</code> $\rightarrow$ <i>real</i> Real (Gas)	

electric

<code>\rms</code> $\rightarrow$ <i>RMS</i> Root Mean Square	<code>\dc</code> $\rightarrow$ <i>DC</i> DC	<code>\ac</code> $\rightarrow$ <i>AC</i> AC
<code>\RMS</code> $\rightarrow$ RMS RMS (up)	<code>\DC</code> $\rightarrow$ DC DC (up)	<code>\AC</code> $\rightarrow$ AC AC (up)
<code>\nom</code> $\rightarrow$ <i>nom</i> Nominal	<code>\ctrl</code> $\rightarrow$ <i>ctrl</i> Control	<code>\meas</code> $\rightarrow$ <i>meas</i> Measured
<code>\setp</code> $\rightarrow$ <i>set</i> Setpoint	<code>\err</code> $\rightarrow$ <i>err</i> Error	<code>\gain</code> $\rightarrow$ <i>gain</i> Gain

2 PGF and TikZ

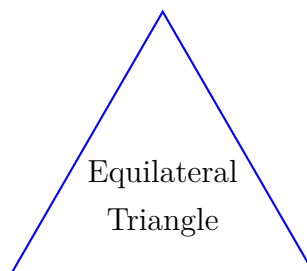
PGF ("Portable Graphics Format") and TikZ ("TikZ ist kein Zeichenprogramm") are powerful tools used to create high-quality diagrams directly within LaTeX.

TikZ is the user-facing layer that provides an intuitive syntax to create graphics. PGF is the lower-level layer responsible for the rendering. Most users interact primarily with TikZ, which internally compiles down to PGF commands. In short:

- TikZ is easier to write and read.
- PGF handles the rendering and offers more fine-grained control (often used via libraries like `pgfplots`).

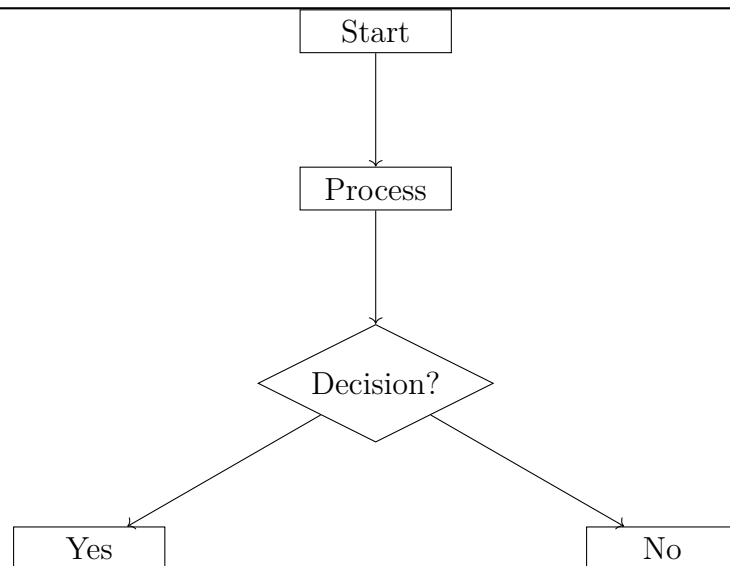
2.1 Baby's first `tikzpicture`

This example draws a simple equilateral triangle using absolute coordinates. It shows how to connect points with lines and place text within the diagram. This kind of drawing is useful for basic geometric figures or quick sketches.



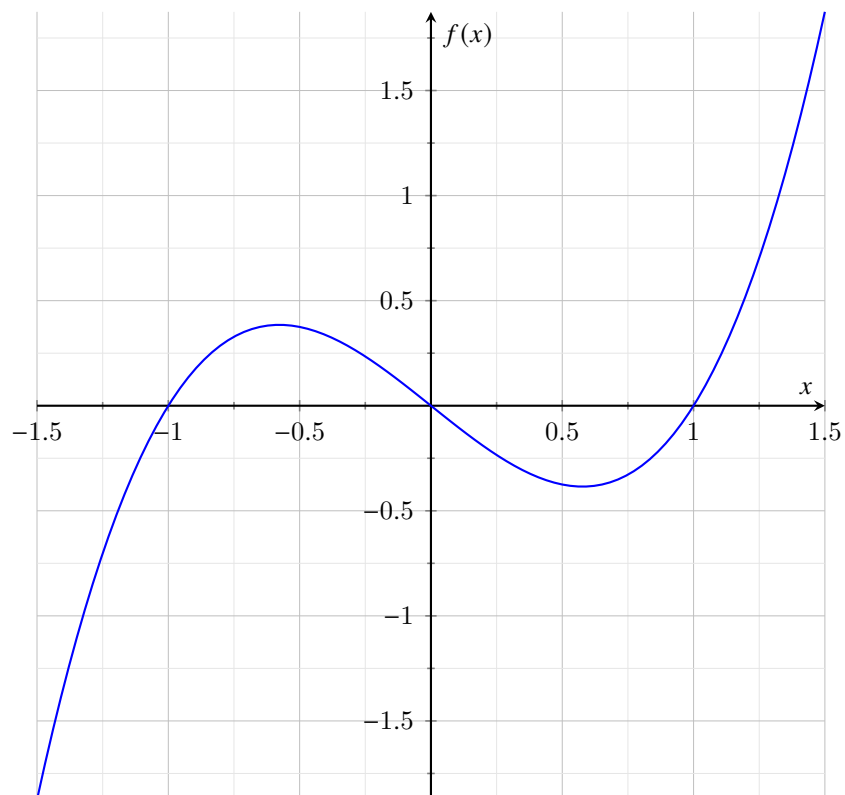
2.2 Flowchart with TikZ

Here we use named nodes and relative positioning to create a simple flowchart. This style makes it easier to maintain and adjust diagrams without dealing with raw coordinates. Shapes like diamonds (used here for decisions) are part of TikZ's geometric shapes library.



2.3 Diagrams with PGFPlots

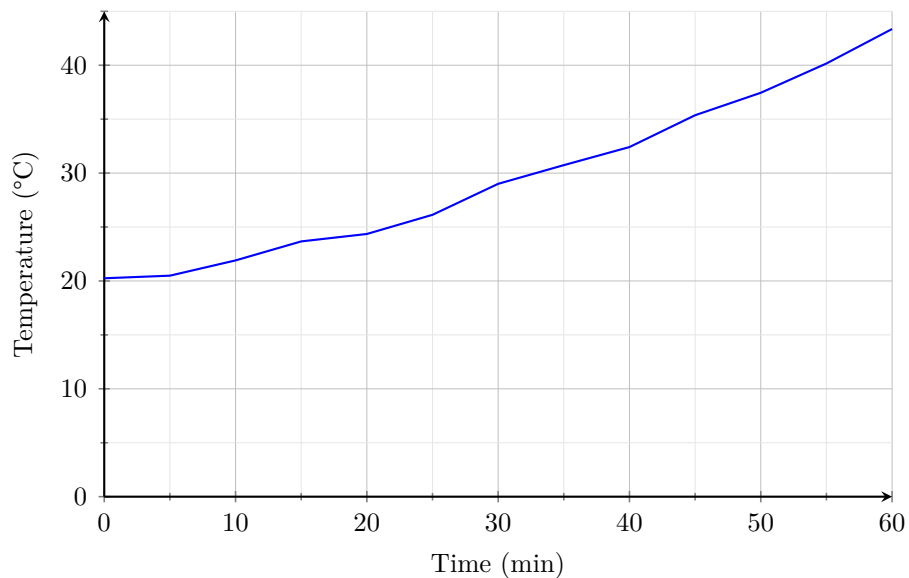
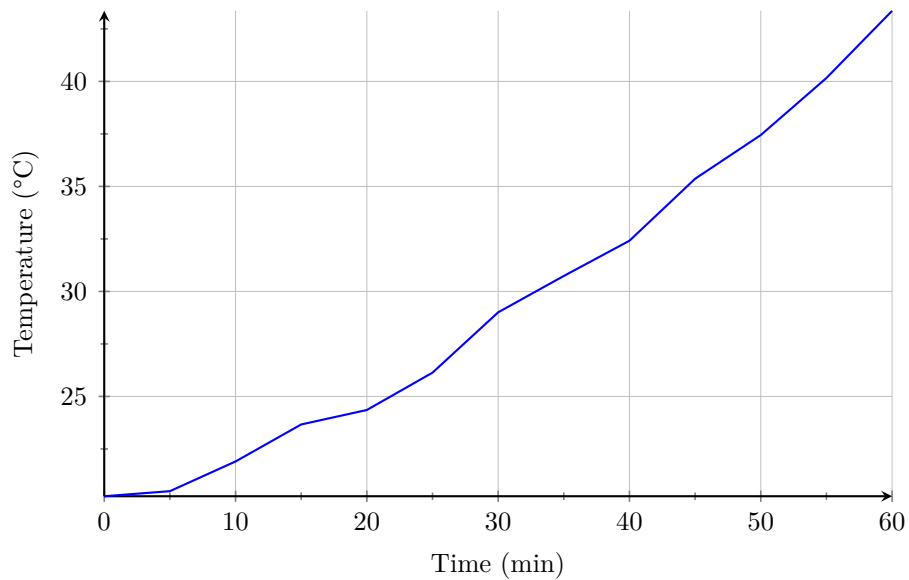
This example uses the `pgfplots` package (built on top of PGF) to plot a mathematical function. PGFPlots is ideal for data visualization and produces publication-quality plots. This one handles scaling, grid lines, and sampling automatically.



2.4 Plotting External Data

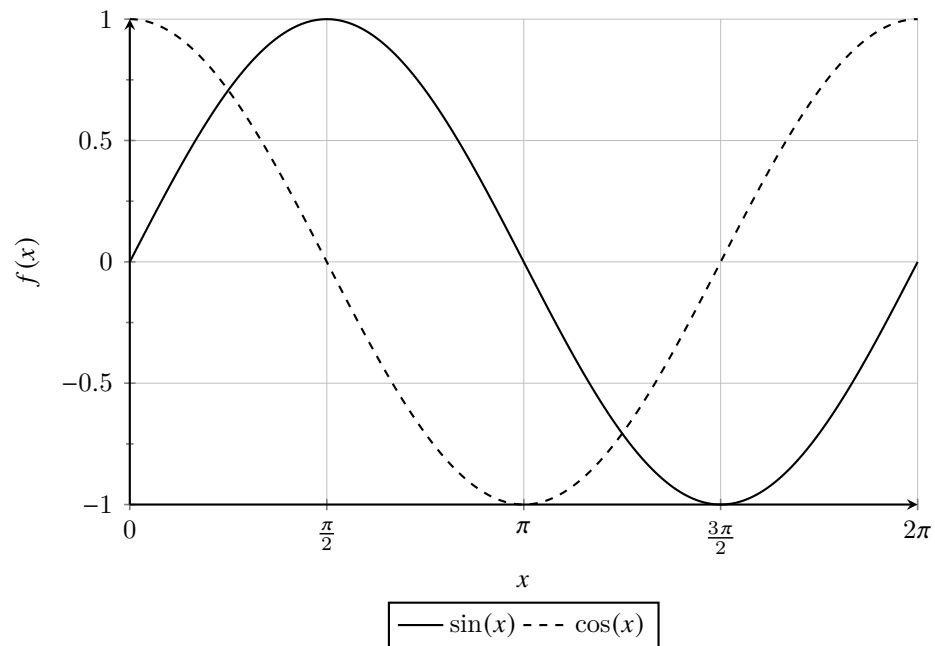
PGFPlots also allows importing data from external files such as CSVs, which is common in scientific documents. Below is a plot of temperature readings taken during a 1-hour experiment at 5-minute intervals.

Make sure the CSV file is accessible and contains headers matching those referenced in the plot code.



2.5 Distinct Line Styles

In academic or black-and-white printing, it is often necessary to distinguish plot lines using different styles (e.g., dashed, dotted) instead of color. This is particularly useful for accessibility or publication constraints. Below is an example showing $\sin(x)$ and $\cos(x)$ using distinct dash patterns.



2.6 Using PGF for tables

PGF (Portable Graphics Format) and its higher-level interface TikZ can be used to create tables in LaTeX. However, PGF/TikZ is primarily designed for graphics and diagrams, so it's not the most common method for making tables.

The `pgfplotstable` packages makes it possible to create tables with PGF. It uses CSV syntax and does not understand strings (letters and words) by default, keep that in mind.

It is not meant to be used for short tables, but more for CSV files and large amount of data.

Tab. 2.1: Example tables utilizing PGF

Name	Age	Score
Alice	23	88
Bob	25	92
Charlie	22	85

Name	Age	Score
Alice	23	88
Bob	25	92
Charlie	22	85

Name	Age	Score
Alice	23	88
Bob	25	92
Charlie	22	85

Here you also see the different rules provided by `booktabs`. Notice that the vertical rules break the horizontal lines. This can be avoided by using custom commands provided. E.g. `\topruled` instead of `\toprule`, the same goes for `\midruled` and `\bottomruled`.

3 Theoretical observation

[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]

4 Bibliography

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- [10] Matthias Schulz. „Doku-Norm in der Praxis“. In: *technische kommunikation* 36.6 (2014), pp. 46–49.

5 List of Figures

1.1	Example of inserting a figure	11
1.2	Example of inserting two figures	11
1.3	Example of inserting a figure with differently sized subfigures with subcaption . . .	12
1.4	Example using the overpic package	13
1.5	A wrapped figure with reduced spacing.	14

6 List of Tables

1.1	Simple Table with Rulings	6
1.2	Table with alternating row colours	6
1.3	Table with alternating colours and automatic column width (B) using tabularx, fixed size (C) and standard centred (A)	7
1.4	Example using tabulary, smaller text to save space	7
1.5	Table from an imported CSV	8
1.6	Longtable from a CSV file	9
1.7	All dimensionless numbers provided	19
2.1	Example tables utilizing PGF	25

A Included PDF files

Currently there is no way to disable a new page from being generated when including a PDF.

FH-Wels-Thesis-Template

This is a re-created template for writing scientific texts and theses at the University of Applied Sciences Upper Austria - Campus Wels. It may be freely used by anyone outside of that institution as well.

Deviation from these instructions is permitted, but support will not be provided for non-standard modifications.

Features

Improvements over the provided version from FHOOE:

- Faster compilation
 - Particularly beneficial for Overleaf (free tier has a 20s compile limit)
- Backwards compatibility with the provided template (to be verified)
- Default TikZ support (TODO)
 - Template for continuous plots (TODO)
 - Template for discrete plots (TODO)

Prerequisites

This template requires the following software:

- TeX Live
- TeXstudio
- GitHub Desktop

To effectively use this template, familiarity with TeX/LaTeX is necessary. Basic knowledge of Git and KOMA-Script is recommended. Many solutions found online may not work correctly with KOMA-Script.

When seeking AI-generated assistance (e.g., ChatGPT, Copilot), specify that you are using KOMA-Script or the `scrbook` document class. As of February 2025, GitHub Copilot with the Claude Sonnet Model is recommended for LaTeX help.

TeX Live is preferred over MiKTeX due to faster compilation and broader package support. This guide assumes a Windows system; official support for Linux and macOS is not provided.

For German users, this [YouTube Playlist](#) serves as a reference. The recommended TeXstudio settings align with those in the playlist, but follow the installation instructions here instead. Whether to use `de_DE`, `de_AT`, or another dictionaries in TeXstudio is up to the user. If dictionaries are unavailable, check [this resource](#).

See the [Extended Playlist](#) for further guidance.

Links to English TeX resources will be added soon.

Installation

Installation time varies, but plan for at least three hours. Avoid using university WiFi; install at home instead. All software must be installed in English.

TeX Live

Download the [TeX Live installer](#) and run it. If prompted with "Windows protected your PC," click `More info`, then `Run Anyway`.

In the TeX Live installer:

1. Close both windows
2. Click the Windows icon and search for `TLSHELL TeX Live Manager`; (see tip below for alternatives)
3. Open it and wait for it to establish a connection
4. Click `All` (should show many packages)
5. Click `Not installed` (should show no packages; otherwise, install all)
6. Click `Updateable` (should show no packages; otherwise, update all)
7. Close the manager

[tip] If `TLSHELL` does not appear in Windows search, open `cmd` and type `tlshell`. Alternatively, navigate to `C:\texlive\2024\bin\windows\tlshell.exe` (adjust year if needed) and create a desktop shortcut.

MiKTeX

TeX Live is recommended, but MiKTeX can be used if preferred. Note that TeX Live compiles 20-30% faster and has better package support.

[!warning] Do not install more than one TeX distribution.

TeXstudio

Installation options:

- Open PowerShell (preferably as admin) and run:

```
winget install texstudio.texstudio --locale en-US
```

- Alternatively, download from [texstudio.org](#) and install manually.

Set the language to English. Other settings can remain at default.

GitHub Desktop

Installation options:

- Use PowerShell (preferably as admin) and run:

```
winget install github.githubdesktop --locale en-US
```

- Alternatively, download from github.com and install manually.

Set the language to English.

Configuration

GitHub Desktop

1. Create a [GitHub account](#) (use a private email, not a university/work address).
2. In GitHub Desktop:
3. Visit the [GitHub repository](#) or scroll to the top
4. Click `<> Code > Open with GitHub Desktop`
5. Click allow in the browser pop
6. A popup in GitHub Desktop will open
7. Set the local path to `C:\Github\[reponame]`

[Important] Do not use long directory paths or OneDrive folders to avoid syncing issues. Use a separate backup solution (e.g., FreeFileSync) for replication to a cloud folder or external storage.

To manually download the template:

1. Visit the [GitHub repository](#)
2. Click `<> Code > Download ZIP`
3. Extract to `C:\Github\[reponame]`

Ensure the file structure is not nested:

```
C:\Github\[reponame]\thesis.tex
```

Delete `README.md`, `LICENSE.md`, and `.gitignore` if not using Git.

TeXstudio Configuration

Run `Biber` under `Tools > Commands > Biber`.

Using Git and GitHub Desktop

- Make small, documented commits
- Commit selectively, especially for non-text elements (plots, graphics, tables)
- Before modifying anything, create a branch for the original template
- Create a new branch for writing your thesis
- Leave the main branch as a reference
- If you rename the `thesis.pdf`, make sure to adjust the `.gitignore` file
 - Git and binary files (here: PDFs) do not mix

Using the Template

The main file is `thesis.tex`. Set it as root manually or by opening `thesis.txss2` from GitHub Desktop or File Explorer.

The template is modular for faster compilation. Disable unused modules by commenting out `\input{modulename}` under `MODULES`.

Use `\include{}` for chapters and sections. Use `\input` for everything else, it is a "dumb" Use `\includeonly{}` to only compile selected sections for faster compiles.

First, compile the template. Check the `Log` below for errors, warnings, or info messages. Save a copy of `thesis.pdf` as `template.pdf` for reference.

First Steps

Data Entry

Edit `preamble/titlepage-data.tex` with relevant details.

Digital Signature

Enable/disable the digital signature in `preamble/signature.tex` by following the instructions in the comments.

Language Selection

Set language after `\begin{document}` :

- Use `\selectlanguage{}` (options: `english`, `ngerman`, `naustrian`)
- Choose `\titlepageDE` or `\titlepageEN`
- Select `\input{preamble/declarationDE}` or `\input{preamble/declarationEN}`
- Set the locale for `\sisetup`
- Provide both English and German abstracts

[!note] Do not delete template code; comment it out instead.

B Source Code

Please be aware that the indentations in this code will be visible in the compiled PDF. Many styles are predefined for you, change them in `preamble/code.tex` if needed.

B.1 Markup and data languages

B.1.1 L^AT_EX code example

```
\documentclass[
  onside,
  bibliography=totocnumbered, % Include bibliography in TOC with number
  listof=totocnumbered,      % Include List of Tables/Figures in TOC with
    ...number
5  numbers=noenddot           % Changes heading numbering from 1.1. to 1.1
  % appendixprefix = true     % Makes Appendix A... First appendix chapter,
    ...disable for one appendix
]{scrbook}
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
\begin{document}
10  Hello, world!

  \begin{equation}
    \pdv{\psi}{t}
    =
15    \frac{\partial \psi}{\partial t}
  \end{equation}

\end{document}
```

B.1.2 YAML code example

You can also directly input the code of a file into the `lstlisting` command.

```
version: '3.8'

services:
  app:
5    image: node:18
    ports:
      - "3000:3000"

  db:
10   image: postgres:15
    environment:
      POSTGRES_USER: user
      POSTGRES_PASSWORD: password
```

B.2 Programming languages

B.2.1 Python code example

```

def fibonacci(n):
    a, b = 0, 1
    while a < n:
        print(a, end=" ")
        a, b = b, a + b
    print()

# Example usage
fibonacci(50)

```

B.2.2 Matlab code example

```

function fibonacci(n)
    a = 0;
    b = 1;
    while a < n
        fprintf('%d ', a);
        temp = a + b;
        a = b;
        b = temp;
    end
    fprintf('\n');
end

% Example usage
fibonacci(50)

```

B.2.3 Wolfram Mathematica code example

```

FibonacciSequence[n_] := Module[{a = 0, b = 1, temp},
    While[a < n,
        Print[a, " "];
        temp = a + b;
        a = b;
        b = temp;
    ]
]

(* Example usage *)
FibonacciSequence[50]

```

B.2.4 C code example

```

#include <stdio.h>

void fibonacci(int n) {
    int a = 0, b = 1, temp;
    while (a < n) {
        printf("%d ", a);
        temp = a + b;
        a = b;
        b = temp;
    }
    printf("\n");
}

```

```
15  int main() {  
    fibonacci(50);  
    return 0;  
}
```

B.2.5 Longer code example (C)

The following is the main.cpp from the TeXstudio GitHub repository. Direct Link:

<https://github.com/texstudio-org/texstudio/blob/master/src/main.cpp>

```

/*****
 *   copyright      : (C) 2003-2007 by Pascal Brachet
 *   addons by Frederic Devernay <frederic.devernay@m4x.org>
 *   addons by Luis Silvestre
5  *   http://www.xmlmath.net/texmaker/
 *
 *   This program is free software; you can redistribute it and/or modify
 *   it under the terms of the GNU General Public License as published by
 *   the Free Software Foundation; either version 2 of the License, or
10  *   (at your option) any later version.
 *
 *****/

#include "mostQtHeaders.h"
15 /*! \mainpage TexStudio
 *
 * \see Texstudio
 * \see PDFDocument
 */

20
#include "texstudio.h"
#include "smallUsefulFunctions.h"
#include "debughelper.h"
#include "debuglogger.h"
25 #include "utilsVersion.h"
#include <qtsingleapplication.h>
#include <QSplashScreen>

#ifdef Q_OS_WIN32
30 #include "windows.h"
typedef BOOL (WINAPI *AllowSetForegroundWindowFunc)(DWORD);
#endif

class TexstudioApp : public QtSingleApplication
35 {
public:
    bool initialized;
    QString delayedFileLoad;
    Texstudio *mw; // Moved from private:
40 TexstudioApp(int &argc, char **argv);
    TexstudioApp(QString &id, int &argc, char **argv);
    ~TexstudioApp();
    void init(QStringList &cmdLine); // This function does all the
        ...initialization instead of the constructor.
    /*bool notify(QObject* obj, QEvent* event){
45 //really slow global event logging:
//qWarning(qPrintable(QString("%1 obj %2 named %3 typed %4 child of %5
        ...received %6").arg(QTime::currentTime().toString("HH:mm:ss:zzz")).arg((
        ...long)obj,8,16).arg(obj->objectName()).arg(obj->metaObject()->className
        ...()).arg(obj->parent()?obj->parent()->metaObject()->className():").arg
        ...(event->type())));
        try {
            return QApplication::notify(obj,event);
        } catch (const std::exception& e){
50         qDebug() << "Caught exception: " << e.what();

```

```

        return false;
    }
}*/

55
protected:
    bool event(QEvent *event);
};

60 TextstudioApp::TextstudioApp(int &argc, char **argv) : QtSingleApplication(argc,
    ...argv)
{
    mw = nullptr;
    initialized = false;
}

65
TextstudioApp::TextstudioApp(QString &id, int &argc, char **argv) :
    ...QtSingleApplication(id, argc, argv)
{
    mw = nullptr;
    initialized = false;
70 }

void TextstudioApp::init(QStringList &cmdLine)
{
    QPixmap pixmap(":/images/splash.png");
75    QSplashScreen *splash = new QSplashScreen(pixmap);
    splash->show();
    processEvents();

    mw = new Textstudio(nullptr, Qt::WindowFlags(), splash);
80    connect(this, SIGNAL(lastWindowClosed()), this, SLOT(quit()));
    splash->finish(mw);
    delete splash;

    initialized = true;

85
    if (!delayedFileLoad.isEmpty()) cmdLine << delayedFileLoad;
    mw->executeCommandLine(cmdLine, true);
    if (!cmdLine.contains("--auto-tests")){
        mw->startupCompleted();
90    }
}

TextstudioApp::~TextstudioApp()
{
95    delete mw;
}

bool TextstudioApp::event(QEvent *event)
{
100    if (event->type() == QEvent::FileOpen) {
        QFileOpenEvent *oe = static_cast<QFileOpenEvent *>(event);
        if (initialized) mw->load(oe->file());
        else delayedFileLoad = oe->file();
        event->accept();
105        return true;
    }
    return QApplication::event(event);
}

```

```

110 QString generateAppId()
{
    QProcessEnvironment env = QProcessEnvironment::systemEnvironment();
    QString user = env.value("USER");
    if (user.isEmpty()) {
115         user = env.value("USERNAME");
    }
    return QString("%1_%2").arg(TEXTSTUDIO, user);
}

120 QStringList parseArguments(const QStringList &args, bool &outStartAlways)
{
    QStringList cmdLine;
    for (int i = 1; i < args.count(); ++i) {
        QString cmdArgument = args[i];

125
        if (cmdArgument.startsWith('-')) {
            // various commands
            if (cmdArgument == "--start-always")
                outStartAlways = true;
130         else if (cmdArgument == "--no-session")
            ConfigManager::dontRestoreSession = true;
        else if ((cmdArgument == "-line" || cmdArgument == "--line") && (++i <
            ...args.count()))
            cmdLine << "--line" << args[i];
        else if ((cmdArgument == "-page" || cmdArgument == "--page") && (++i <
            ...args.count()))
135         cmdLine << "--page" << args[i];
        else if ((cmdArgument == "-insert-cite" || cmdArgument == "--insert-cite")
            ... && (++i < args.count()))
            cmdLine << "--insert-cite" << args[i];
        else if (cmdArgument == "--ini-file" && (++i < args.count())) {
            // deprecated: use --config instead
140         ConfigManager::configDirOverride = QFileInfo(args[i]).absolutePath();
        }
        else if (cmdArgument == "--config" && (++i < args.count()))
            ConfigManager::configDirOverride = args[i];
#ifdef DEBUG_LOGGER
145         else if ((cmdArgument == "--debug-logfile") && (++i < args.count()))
            debugLoggerStart(args[i]);
        #endif
        else
            cmdLine << cmdArgument;
150     } else {
        if (cmdArgument.endsWith(".txss") || cmdArgument.endsWith(".txss2")) {
            // explicit session restor
            // disable restoring of last session
            ConfigManager::dontRestoreSession = true;
155         }
        cmdLine << QFileInfo(cmdArgument).absoluteFilePath();
    }
}

return cmdLine;
160 }

bool handleCommandLineOnly(const QStringList &cmdLine) {
    // note: stdout is not supported for Win GUI applications. Will simply not
    ...output anything there.
    if (cmdLine.contains("--help")) {

```



```

165   QTextStream(stdout) << "Usage: texstudio [options] [file]\n"
      << "\n"
      << "Options:\n"
      << "  --config DIR           use the specified settings directory\n"
      << "  --master              define the document as explicit root
      ...document\n"
170   << "  --line LINE[:COL]    position the cursor at line LINE and column
      ...COL\n"
      << "  --insert-cite CITATION inserts the given citation\n"
      << "  --start-always       start a new instance, even if TXS is already
      ... running\n"
      << "  --pdf-viewer-only    run as a standalone pdf viewer without an
      ...editor\n"
      << "  --page PAGENUM       display a certain page in the pdf viewer\n"
175   << "  --no-session        do not load/save the session at startup/
      ...close\n"
      << "  --texpath PATH       force resetting command defaults with PATH
      ...as first search path\n"
      << "  --version           show version number\n"
      #ifdef DEBUG_LOGGER
      << "  --debug-logfile pathname write debug messages to pathname\n"
180   #endif
      ;
      return true;
  }

185   if (cmdLine.contains("--version")) {
      QTextStream(stdout) << "TeXstudio " << TXSVERSION << " (" <<
      ...TEXSTUDIO_GIT_REVISION << ")\n";
      return true;
  }

190   return false;
}

int main(int argc, char **argv)
{
195   QString appId = generateAppId();
      #if QT_VERSION >= QT_VERSION_CHECK(5,6,0)
      if(qEnvironmentVariableIntValue("TEXSTUDIO_HIDPI_SCALE")>0){
          QApplication::setAttribute(Qt::AA_EnableHighDpiScaling);
      } else {
200   QApplication::setAttribute(Qt::AA_DisableHighDpiScaling);
      }
      #endif
      // This is a dummy constructor so that the programs loads fast.
      TexstudioApp a(appId, argc, argv);
205   bool startAlways = false;
      QStringList cmdLine = parseArguments(QCoreApplication::arguments(),
      ...startAlways);

      if (handleCommandLineOnly(cmdLine)) {
          return 0;
210   }

      if (!startAlways) {
          if (a.isRunning()) {
              #ifdef Q_OS_WIN32
215   AllowSetForegroundWindowFunc asfw = (AllowSetForegroundWindowFunc)
              ...GetProcAddress(GetModuleHandleA("user32.dll"), "

```

```
        ...AllowSetForegroundWindow");
        if (asfw) asfw(/*ASFW_ANY*/(DWORD)(-1));
        #endif
        a.sendMessage(cmdLine.join("#!#"));
        return 0;
220    }
    }

    a.setApplicationName( TEXTSTUDIO );
    #if (QT_VERSION >= QT_VERSION_CHECK(5, 7, 0)) && defined(Q_OS_LINUX)
225    a.setDesktopFileName("textstudio");
    #endif
    a.init(cmdLine); // Initialization takes place only if there is no other
        ...instance running.

    QObject::connect(&a, SIGNAL(messageReceived(const QString&)),
230    a.mw, SLOT(onOtherInstanceMessage(const QString&)));

    try {
        int execResult = a.exec();
        #ifdef DEBUG_LOGGER
235        if (debugLoggerIsLogging()) {
            debugLoggerStop();
        }
        #endif
        return execResult;
240    } catch (...) {
        #ifndef NO_CRASH_HANDLER
            catchUnhandledException();
        #endif
        throw;
245    }
    }
```